NPTEL — Programming in Java

All 150 Questions — Grouped by Topic

Course: noc26_cs36 | Mock Exams 1–5

Format: Answer key appears after each topic group for quick self-evaluation.

Topic 1: Java Basics & History

M1-Q10

Q1. According to the textbook, who is credited as the pioneer of the team that first introduced the concepts leading to Java?

A) Debasis Samanta
B) Patrick Naughton
C) James Gosling
D) Mike Sheridan

M2-Q10

Q2. According to the textbook's introduction, which team at Sun Microsystems initially developed the programming language that would later become Java?

A) The Blue Team
B) The Green Team
C) The Oak Team
D) The Netscape Team

M3-Q10

Q3. According to the textbook's introduction, who was the pioneer of the 'Green Team' that initiated the project which eventually led to Java?

A) Debasis Samanta
B) Bill Gates
C) James Gosling
D) Linus Torvalds

M4-Q10

Q4. Which of the following is a primary characteristic of function-oriented programming that contrasts with OOP?

A) Data and functions bundled in objects
B) Bottom-up design approach
C) Data is often shared globally among a large set of small functions
D) Inherently supports polymorphism and inheritance

M5-Q10

Q5. According to the text, who is considered the pioneer of the 'Green Team' that first introduced the concept of Java?

A) Debasis Samanta
B) James Gosling
C) Bill Gates
D) Linus Torvalds

M1-Q13

Q6. Which of the following is NOT listed as a feature of the Java programming language?

A) Portable
B) Secure
C) High Performance
D) Procedural

M2-Q16

Q7. Which of the following is NOT one of the four main paradigms of OOP?

A) Encapsulation
B) Inheritance

C) Recursion
D) Polymorphism

■ **Answer Key — Java Basics & History**

Q1 (M1-Q10): C	Q2 (M2-Q10): B	Q3 (M3-Q10): C
Q4 (M4-Q10): C	Q5 (M5-Q10): B	Q6 (M1-Q13): D
Q7 (M2-Q16): C		

Topic 2: Java Tools & Commands (javac, java, jdb, javadoc)

M1-Q2

Q1. Which JDK tool is a command-line debugger used for finding and fixing bugs in Java programs?

A) javac
B) javadoc
C) jdb
D) jre

M1-Q11

Q2. What is the correct command to compile a Java source file named 'HelloWorldApp.java'?

A) java HelloWorldApp.java
B) javac HelloWorldApp.java
C) run HelloWorldApp.java
D) compile HelloWorldApp

M1-Q14

Q3. Which JDK tool is used for generating API documentation from source code comments?

A) javac
B) javap
C) javadoc
D) jdb

M2-Q2

Q4. Which Java command-line tool generates API documentation in HTML format?

A) java
B) javac
C) jdb
D) javadoc

M2-Q11

Q5. What is the correct command to compile 'HelloWorldApp.java'?

A) java HelloWorldApp
B) javac HelloWorldApp.java
C) run HelloWorldApp.java
D) java HelloWorldApp.class

M2-Q18

Q6. Which Java tool generates API documentation from comments in source code?

A) javac
B) javadoc
C) javap

D) javah

M3-Q2

Q7. Which Java command-line tool generates API documentation in HTML format?

A) javac
B) java
C) javadoc
D) jdb

M3-Q12

Q8. Which JDK tool compiles a .java source file into a .class bytecode file?

A) java
B) javadoc
C) javac
D) javap

M4-Q1

Q9. Which command compiles 'MyProgram.java' into bytecode?

A) java MyProgram.java
B) run MyProgram.java
C) javac MyProgram.java
D) jdb MyProgram

M4-Q11

Q10. What is the command used to compile 'MyProgram.java'?

A) java MyProgram
B) javac MyProgram.java
C) run MyProgram.java
D) compile MyProgram

M5-Q1

Q11. Which Java tool finds/fixes bugs by allowing step-by-step execution?

A) javac
B) javadoc
C) java
D) jdb

M5-Q17

Q12. Which JDK tool generates API-style documentation from source code comments?

A) javac
B) javap
C) javadoc
D) javah

■ Answer Key — Java Tools & Commands (javac, java, jdb, javadoc)

Q1 (M1-Q2): C	Q2 (M1-Q11): B	Q3 (M1-Q14): C
Q4 (M2-Q2): D	Q5 (M2-Q11): B	Q6 (M2-Q18): B
Q7 (M3-Q2): C	Q8 (M3-Q12): C	Q9 (M4-Q1): C
Q10 (M4-Q11): B	Q11 (M5-Q1): D	Q12 (M5-Q17): C

Topic 3: Java Programming Steps & Bytecode

M1-Q15

Q1. What is the standard file extension for a compiled Java class file?

A) .java
B) .class
C) .obj
D) .exe

M1-Q21

Q2. How is platform independence primarily achieved in Java?

A) Different compiler for each OS
B) Compiling to machine-specific native code
C) Compiling to bytecode (.class file) run on any JVM
D) Using pointers across platforms

M2-Q1

Q3. What is compiled Java code called that can be executed by the JVM?

A) Machine Code
B) Source Code
C) Bytecode
D) Opcode

M2-Q15

Q4. What is the file extension for a Java source code file?

A) .class
B) .exe
C) .obj
D) .java

M3-Q11

Q5. What is the file extension for a compiled Java program containing bytecode?

A) .java
B) .exe
C) .class
D) .obj

M3-Q17

Q6. What key feature of Java compilation enables platform independence?

A) Compiles to native machine code
B) Compiles to intermediate bytecode executed by JVM
C) Requires separate compiler per OS
D) Source bundled with compiled program

M4-Q17

Q7. What is the key component produced by javac that enables platform independence?

A) Native executable (.exe)
B) Platform-specific library
C) Source map file
D) .class file with platform-neutral bytecode

M5-Q11

Q8. What is the correct file extension for a Java source code file?

A) .class
B) .exe
C) .java
D) .j

■ Answer Key — Java Programming Steps & Bytecode

Q1 (M1-Q15): B	Q2 (M1-Q21): C	Q3 (M2-Q1): C
Q4 (M2-Q15): D	Q5 (M3-Q11): C	Q6 (M3-Q17): B
Q7 (M4-Q17): D	Q8 (M5-Q11): C	

Topic 4: Keywords & Reserved Words

M1-Q1

Q1. Which of the following is a reserved keyword in Java?

A) main
B) String
C) case
D) malloc

M2-Q3

Q2. Which of the following cannot be used as a variable name in Java?

A) string
B) main
C) case
D) System

M3-Q1

Q3. Which of the following is a reserved keyword in Java?

A) string
B) main
C) System
D) case

■ Answer Key — Keywords & Reserved Words

Q1 (M1-Q1): C	Q2 (M2-Q3): C	Q3 (M3-Q1): D

Topic 5: OOP Concepts — Encapsulation, Polymorphism, Inheritance

M1-Q12

Q1. The concept of bundling data and methods into a single unit is known as:

A) Inheritance
B) Polymorphism
C) Encapsulation
D) Information Hiding

M1-Q16

Q2. In Java inheritance, what is another common term for a 'superclass'?

A) Child class
B) Derived class

C) Parent class
D) Subclass

M1-Q18

Q3. Which statement best describes the OOP principle of polymorphism?

A) Ability to derive new class from existing
B) Ability of a method with same name to perform different actions
C) Bundling data and methods
D) Restricting access to class members

M2-Q12

Q4. Bundling data and methods that operate on that data into a single unit is known as:

A) Inheritance
B) Polymorphism
C) Encapsulation
D) Information Hiding

M2-Q13

Q5. Which keyword is used to create a subclass in Java?

A) inherits
B) extends
C) super
D) implements

M2-Q26

Q6. What is the outcome of this dynamic method dispatch code? Animal a = new Dog(); a.run();

A) Fails to compile
B) 'Animal is running'
C) 'Dog is running'
D) Runtime error

M3-Q13

Q7. What is the term for bundling data (fields) and methods into a single unit like a class?

A) Inheritance
B) Polymorphism
C) Encapsulation
D) Abstraction

M3-Q15

Q8. Which Java keyword indicates a class is inheriting from a superclass?

A) inherits
B) implements
C) extends
D) super

M3-Q21

Q9. When a subclass defines a method with same name, return type, and parameters as superclass, this is:

A) Method overloading
B) Method overriding
C) Method hiding
D) Constructor chaining

M3-Q27

Q10. Bike b = new Splendor(); b.run(); — which run() is called?

A) Bike.run() — reference is Bike
B) Splendor.run() — compile-time binding
C) Fails to compile
D) Splendor.run() — runtime polymorphism

M4-Q14

Q11. Which keyword creates a subclass that inherits from a superclass?

A) extends
B) implements
C) inherits
D) super

M4-Q20

Q12. Which statement correctly describes method overriding in Java?

A) Subclass defines multiple methods with same name but different params
B) Superclass prevents subclass from defining same method
C) Subclass provides specific implementation for method already in superclass
D) Class implements method from interface

M5-Q13

Q13. What Java keyword is used to specify that a class is inheriting from another class?

A) inherits
B) implements
C) extends
D) super

M5-Q22

Q14. When a method in a subclass has the same name, return type, and parameters as the superclass, this is:

A) Method Overloading
B) Method Overriding
C) Method Hiding
D) Method Finalizing

■ Answer Key — OOP Concepts — Encapsulation, Polymorphism, Inheritance

Q1 (M1-Q12): C	Q2 (M1-Q16): C	Q3 (M1-Q18): B
Q4 (M2-Q12): C	Q5 (M2-Q13): B	Q6 (M2-Q26): C
Q7 (M3-Q13): C	Q8 (M3-Q15): C	Q9 (M3-Q21): B
Q10 (M3-Q27): D	Q11 (M4-Q14): A	Q12 (M4-Q20): C
Q13 (M5-Q13): C	Q14 (M5-Q22): B	

Topic 6: Static Keyword & Instance vs Class Variables

M1-Q17

Q1. What is the primary difference between an instance variable and a class variable (static)?

A) Instance variables shared by all objects
B) Class variables shared by all objects, each object has own instance variable
C) Only class variables can be final

D) Class variables not accessible by instance methods

M1-Q28

Q2. If a method is declared as static, what is the implication for calling it?

A) Only from within same class
B) Object must be created first
C) Can be called without creating an object
D) Only by other static methods

M2-Q17

Q3. In main(String args[]), what does the static keyword indicate?

A) Method returns no value
B) Accessible from any class
C) Can be called without creating an object
D) Method value is constant

M2-Q28

Q4. If a variable is declared with static, what does it signify?

A) Value cannot change
B) Each object gets own copy
C) Associated with class, shared by all objects
D) Only same-class methods can access it

M3-Q20

Q5. What is the fundamental difference between instance variable and class variable (static)?

A) Instance variables are final
B) Each object has own instance variable; class variable shared among all
C) Class variables must be initialized at declaration
D) Only class variables accessed by static methods

M3-Q30

Q6. If a class has static int classCounter=0 and non-static int instanceId, and 5 instances created — how many copies of each exist?

A) 5 and 1
B) 1 and 5
C) 5 and 5
D) 1 and 1

M4-Q19

Q7. What is the key difference between instance variable and static variable?

A) Static cannot be changed
B) Static has one copy shared among all instances; each instance has own instance variable
C) Static only accessed within its class
D) Instance stored on stack, static on heap

M5-Q20

Q8. The main method must be declared static. Why?

A) To be accessed by subclasses
B) To be called by JVM without creating an object
C) To prevent overriding
D) Accessible only within package

M5-Q21

Q9. Primary distinction between instance variable and static variable?

A) Instance variable shared by all; static has separate copy per object
B) Static shared by all; instance has separate copy per object
C) Instance cannot change after init
D) Static only for primitive types

■ Answer Key — Static Keyword & Instance vs Class Variables

Q1 (M1-Q17): B	Q2 (M1-Q28): C	Q3 (M2-Q17): C
Q4 (M2-Q28): C	Q5 (M3-Q20): B	Q6 (M3-Q30): B
Q7 (M4-Q19): B	Q8 (M5-Q20): B	Q9 (M5-Q21): B

Topic 7: Constructors & this Keyword

M1-Q20

Q1. What is the purpose of a constructor in a Java class?

A) Destroy object and free memory
B) Define main execution point
C) Automatically initialize object when created
D) Define static method

M3-Q26

Q2. In constructor public MyClass(int value) { this.value = value; }, what is the role of this?

A) Creates new instance
B) Calls superclass method
C) Distinguishes instance variable from constructor parameter
D) Declares parameter as final

M4-Q13

Q3. What is the special method that shares the class name and initializes newly created objects?

A) main()
B) init()
C) A constructor
D) new()

M4-Q27

Q4. In this.name = name, what is the primary role of the this keyword?

A) Creates new instance
B) Calls superclass method
C) Refers to static context
D) Disambiguates instance variable from constructor parameter

M5-Q12

Q5. What is the primary purpose of a constructor as described in encapsulation?

A) To destroy an object
B) To automatically initialize an object when created
C) To return a value
D) To define the main entry point

M5-Q19

Q6. What is the term for defining multiple constructors with different parameter lists?

A) Constructor overriding
B) Constructor inheritance

C) Constructor chaining
D) Constructor overloading

M5-Q2

Q7. Which is the correct syntax to declare and instantiate a Vehicle object?

A) Vehicle car = new Vehicle;
B) Vehicle car = new Vehicle();
C) new Vehicle() = car;
D) Vehicle car = Vehicle();

■ Answer Key — Constructors & this Keyword

Q1 (M1-Q20): C	Q2 (M3-Q26): C	Q3 (M4-Q13): C
Q4 (M4-Q27): D	Q5 (M5-Q12): B	Q6 (M5-Q19): D
Q7 (M5-Q2): B		

Topic 8: super Keyword & Inheritance Mechanics

M1-Q22

Q1. What is the purpose of the 'super' keyword in inheritance?

A) Create instance of superclass
B) Declare class as superclass
C) Refer to members of immediate parent class
D) Mark method as non-overridable

M2-Q23

Q2. What is super(arg1, arg2) used for in a subclass constructor?

A) Refer to static method in superclass
B) Create new instance of superclass
C) Call a specific constructor of superclass
D) Access private member of superclass

M3-Q24

Q3. Which keyword explicitly calls a superclass's constructor from a subclass constructor?

A) this()
B) new()
C) extends()
D) super()

M4-Q28

Q4. How would you call the display() method of the immediate superclass if the subclass overrode it?

A) Super.display();
B) this.Super.display();
C) super.display();
D) new Super().display();

M5-Q25

Q5. How can a Dog class constructor invoke a constructor from the Animal superclass?

A) By calling Animal()
B) By calling this()
C) By calling super()

D) Called automatically and cannot be explicitly invoked

M5-Q26

Q6. How can overridden work() in subclass explicitly invoke Employee's version of work()?

A) Not possible once overridden
B) By calling this.work()
C) By calling super.work()
D) By casting to superclass

M1-Q25

Q7. What is the primary reason Java does not support multiple inheritance for classes?

A) To prevent the 'Diamond Problem'
B) Improve runtime performance
C) Interfaces are more efficient
D) Reduce bytecode size

M4-Q24

Q8. Which statement is true regarding Java's support for multiple inheritance?

A) A class can extend multiple classes
B) Can't extend but can implement multiple interfaces
C) No multiple inheritance of classes; but can implement multiple interfaces
D) Multiple inheritance via super keyword

■ Answer Key — super Keyword & Inheritance Mechanics

Q1 (M1-Q22): C	Q2 (M2-Q23): C	Q3 (M3-Q24): D
Q4 (M4-Q28): C	Q5 (M5-Q25): C	Q6 (M5-Q26): C
Q7 (M1-Q25): A	Q8 (M4-Q24): C	

Topic 9: Access Modifiers & Information Hiding

M2-Q19

Q1. Which access modifier provides the most restrictive access?

A) public
B) protected
C) private
D) default

M2-Q25

Q2. A protected method in class A (package P1) is accessible from which of the following?

A) Any class in any package
B) Only from class A
C) Any class in P1, and any subclass of A in any package
D) Only subclasses of A in P1

M3-Q22

Q3. Which access modifier restricts visibility to its own class only?

A) public
B) protected
C) private
D) No modifier

M3-Q28

Q4. Class Derived in package p2 extends Base in package p1. Is protected method doAction() accessible from Derived?

A) No, different package
B) Yes, because Derived is a subclass
C) Only if also in p1
D) Only if Base instance in p2

M4-Q16

Q5. Which access modifier offers least restriction?

A) private
B) protected
C) public
D) default

M4-Q21

Q6. Which access modifier makes a member accessible only within its own class?

A) public
B) private
C) protected
D) default

M4-Q26

Q7. How can SubClass in packageB access protected method calculate() from MyClass in packageA?

A) Impossible
B) By creating MyClass instance within SubClass
C) SubClass must extend MyClass
D) Must import and method must be static

M5-Q14

Q8. Which access modifier offers widest scope?

A) private
B) protected
C) default
D) public

M5-Q27

Q9. Class B in package p2 (not subclass of A). Can B's object access A's protected method?

A) Yes, protected accessible everywhere
B) Yes, if B imports p1
C) No, protected limited to same package or subclasses
D) No, only within same class

M1-Q26

Q10. ClassX in P1 has protected method doWork(). ClassY in P2 extends ClassX. From where is doWork() accessible without a ClassY object?

A) Only from ClassX and subclasses
B) From any class in P1 but not ClassY
C) From any class in P1 only
D) From any class in any package

■ Answer Key — Access Modifiers & Information Hiding

Q1 (M2-Q19): C	Q2 (M2-Q25): C	Q3 (M3-Q22): C

Q4 (M3-Q28): B	Q5 (M4-Q16): C	Q6 (M4-Q21): B
Q7 (M4-Q26): C	Q8 (M5-Q14): D	Q9 (M5-Q27): C
Q10 (M1-Q26): C		

Topic 10: Interfaces & Abstract Classes

M1-Q7

Q1. Which is a valid way to define a Java interface?

A) public interface MyInterface { void myMethod() { ... } }
B) private interface MyInterface { void myMethod(); }
C) public interface MyInterface { default void myMethod() { ... } }
D) public interface MyInterface { protected void myMethod(); }

M1-Q30

Q2. Which statement accurately describes the relationship between abstract class and interface?

A) Abstract class can have instance variables; interface cannot
B) A class can implement multiple abstract classes
C) Interface can be instantiated; abstract class cannot
D) All methods in abstract class must be abstract

M2-Q20

Q3. What is the primary purpose of an interface in Java?

A) To create concrete objects
B) To define a set of methods a class must implement
C) To store instance variables
D) To provide complete method implementations

M2-Q27

Q4. Key difference between interface and abstract class?

A) Interface cannot be implemented, only extended
B) Abstract class can have constructors; interface cannot
C) Class can extend multiple abstract classes but implement only one interface
D) Interface can have instance variables; abstract class cannot

M2-Q29

Q5. A Java class can simultaneously:

A) Extend multiple superclasses and implement multiple interfaces
B) Extend one superclass and implement multiple interfaces
C) Extend multiple superclasses but implement only one interface
D) Extend one superclass and implement at most one interface

M3-Q23

Q6. A class uses implements for _____ and extends for _____:

A) Class, Interface
B) Package, Class
C) Interface, Class
D) Abstract class, Package

M3-Q29

Q7. Why does Java support implementing multiple interfaces but only extending single class?

A) Extending class more expensive

B) Interfaces cannot contain methods
C) To prevent Diamond Problem — interfaces only provide signatures
D) Arbitrary language design

M4-Q7

Q8. Which correctly describes a 'marker' or 'tagged' interface in Java?

A) Contains exactly one method
B) Has no methods or fields; provides runtime type info
C) Only implemented by final classes
D) Contains only static methods

M4-Q22

Q9. Which keyword does a class use to adopt an interface?

A) extends
B) throws
C) uses
D) implements

M4-Q29

Q10. What are the implicit modifiers for a method declared within a Java interface?

A) public and static
B) private and final
C) public and abstract
D) protected and abstract

M5-Q24

Q11. What modifiers are implicitly applied to any variable declared within a Java interface?

A) private, static
B) public, static, final
C) protected, final
D) public, volatile

M5-Q28

Q12. Key difference between interface and abstract class concerning inheritance?

A) Abstract class can implement interface; interface cannot extend abstract class
B) Class can implement multiple interfaces but only extend one abstract class
C) Abstract classes have constructors; interfaces cannot
D) Interface can only have public methods; abstract class can have private

■ Answer Key — Interfaces & Abstract Classes

Q1 (M1-Q7): C	Q2 (M1-Q30): A	Q3 (M2-Q20): B
Q4 (M2-Q27): B	Q5 (M2-Q29): B	Q6 (M3-Q23): C
Q7 (M3-Q29): C	Q8 (M4-Q7): B	Q9 (M4-Q22): D
Q10 (M4-Q29): C	Q11 (M5-Q24): B	Q12 (M5-Q28): B

Topic 11: Packages & import

M1-Q3

Q1. Which package is implicitly imported into every Java program by default?

A) java.util

B) java.io
C) java.net
D) java.lang

M2-Q22

Q2. What is the correct way to import all classes from java.util package?

A) import java.util;
B) include java.util.*;
C) import java.util.all;
D) import java.util.*;

M3-Q6

Q3. Which statement regarding java.lang package is FALSE?

A) Automatically imported
B) String class (final) is part of it
C) Cloneable class is part of it
D) Thread class is part of it

M3-Q16

Q4. What is the primary purpose of packages in Java?

A) Increase execution speed
B) Group related classes, prevent naming conflicts
C) Define main entry point
D) Handle runtime exceptions

M4-Q3

Q5. Packages are primarily used for:

A) Manage memory allocation
B) Enable multiple inheritance
C) Organize related classes and prevent naming conflicts
D) Handle runtime exceptions

M4-Q12

Q6. Which Java package is imported by default into every Java program?

A) java.io
B) java.util
C) java.lang
D) java.awt

M4-Q15

Q7. Primary purpose of packages in Java?

A) Automatically handle exceptions
B) Improve runtime speed
C) Organize classes and avoid naming conflicts
D) Define main method

M5-Q15

Q8. Primary advantage of organizing classes into packages?

A) Increase execution speed
B) Avoid namespace collisions and organize related classes
C) Automatically handle exceptions
D) Reduce compiled code size

M5-Q16

Q9. What does import java.util.*; accomplish?

A) Imports only util class
B) Imports all public classes/interfaces from java.util
C) Creates new package java.util
D) Imports all classes including private ones

M5-Q23

Q10. Which environment variable helps locate user-defined packages?

A) JAVAPATH
B) PATH
C) CLASSPATH
D) JDK_HOME

■ Answer Key — Packages & import

Q1 (M1-Q3): D	Q2 (M2-Q22): D	Q3 (M3-Q6): C
Q4 (M3-Q16): B	Q5 (M4-Q3): C	Q6 (M4-Q12): C
Q7 (M4-Q15): C	Q8 (M5-Q15): B	Q9 (M5-Q16): B
Q10 (M5-Q23): C		

Topic 12: Exception Handling

M2-Q5

Q1. Output of: try{ 10/0 } catch(ArithmeticException e){ print('Catch') } finally{ print('Finally') } print('End')?

A) CatchFinallyEnd
B) TryCatchFinallyEnd
C) Catch
D) Runtime error and program terminates

M3-Q3

Q2. All of the following are keywords in Java's exception handling EXCEPT:

A) try
B) thrown
C) finally
D) catch

M3-Q8

Q3. What is result of this code? catch(Exception e) before catch(ArrayIndexOutOfBoundsException ae)?

A) EF
B) AF
C) A compilation error occurs
D) EAF

M4-Q2

Q4. Which keyword is used to explicitly throw an exception?

A) try
B) throws
C) finally
D) throw

M4-Q6

Q5. What is the rule regarding the order of catch blocks?

A) Subclass exceptions must be caught before superclass
B) Superclass must be caught before subclass
C) Order doesn't matter
D) Only one catch allowed

M5-Q3

Q6. In try-catch-finally, which part is guaranteed to execute?

A) try block
B) catch block
C) finally block
D) Only try and catch

M5-Q8

Q7. Why does this code result in compilation error? catch(Exception) before catch(ArithmeticException)?

A) try cannot have two catch blocks
B) Exception catch must be first
C) finally required with multiple catches
D) ArithmeticException catch unreachable; Exception caught first

■ Answer Key — Exception Handling

Q1 (M2-Q5): A	Q2 (M3-Q3): B	Q3 (M3-Q8): C
Q4 (M4-Q2): D	Q5 (M4-Q6): A	Q6 (M5-Q3): C
Q7 (M5-Q8): D		

Topic 13: Multithreading — Thread vs Runnable, start() vs run()

M1-Q9

Q1. What is the output of a program that calls run() directly instead of start()?

A) 1 3
B) 2 4
C) 1 2 3 4
D) Non-deterministic

M1-Q24

Q2. What is the key difference between Thread class and Runnable interface?

A) Class can extend multiple Thread classes
B) Threads with Runnable cannot be synchronized
C) Implementing Runnable allows inheriting from another class — impossible if extends Thread
D) start() must be manually implemented with Runnable

M2-Q6

Q3. Which statement is true about start() vs run()?

A) Both create new thread
B) run() creates new thread; start() executes in current
C) start() creates new thread; run() executes in current thread
D) Both identical

M3-Q7

Q4. Key difference between calling start() and run() on a Thread object?

A) No difference
B) run() creates new thread; start() executes in current
C) start() creates new thread; run() executes in current
D) run() can be called multiple times to restart

M4-Q9

Q5. A thread in 'dead' state — what happens if start() is called again?

A) Thread restarts
B) Call ignored
C) Program won't compile
D) IllegalThreadStateException thrown

M5-Q5

Q6. Primary difference between invoking start() and run() on a Thread?

A) start() creates new thread; run() executes in current
B) run() creates new thread; start() executes in current
C) Both create new thread but start() has higher priority
D) No difference

■ Answer Key — Multithreading — Thread vs Runnable, start() vs run()

Q1 (M1-Q9): B	Q2 (M1-Q24): C	Q3 (M2-Q6): C
Q4 (M3-Q7): C	Q5 (M4-Q9): D	Q6 (M5-Q5): A

Topic 14: Arrays & Command Line Args

M1-Q6

Q1. Which statement correctly prints the 2nd ('A') and last ('L') characters of array as 'AL'?

A) System.out.print(nptel[1] + nptel[8]);
B) System.out.print("" + nptel[1] + nptel[8]);
C) System.out.print(nptel[2] + nptel[9]);
D) System.out.print(nptel[1] + "" + nptel[8]);

M1-Q23

Q2. Which declares a 2D array of int with 3 rows and 4 columns?

A) int myArray[][] = new int[4][3];
B) int[3][4] myArray = new int[][];
C) int myArray[][] = new int[3][4];
D) int myArray = new int[3, 4];

M1-Q29

Q3. Program executed as: java CommandLineDemo Hello World. Value of args.length?

A) 0
B) 1
C) 2
D) 3

M2-Q8

Q4. java Test 'Java NPTEL' Course 2023. Output of args.length + ' ' + args[0]?

A) 4 Java
B) 3 Java NPTEL

C) 4 Java NPTEL
D) 3 Java

M2-Q9

Q5. System.arraycopy(source, 2, dest, 1, 3) on source={'a','b','c','d','e','f','g'}, dest=new char[5]. Output?

A) [c, d, e, ,]
B) [, c, d, e,]
C) [a, b, c, ,]
D) [, a, b, c,]

M3-Q4

Q6. Which is an invalid declaration for a 2D array?

A) int[][] arr = new int[5][5];
B) int[][] arr = new int[5][];
C) int arr[][] = {{1,2},{3,4}};
D) int[][] arr = new int[5];

M4-Q23

Q7. How to declare integer array 'numbers' and allocate 50 integers?

A) int numbers[50];
B) int numbers[] = new int[50];
C) int numbers = new int(50);
D) Array numbers[50] = new int[];

■ Answer Key — Arrays & Command Line Args

Q1 (M1-Q6): B	Q2 (M1-Q23): C	Q3 (M1-Q29): C
Q4 (M2-Q8): B	Q5 (M2-Q9): B	Q6 (M3-Q4): D
Q7 (M4-Q23): B		

Topic 15: Applet Programming (AWT & Swing)

M1-Q19

Q1. When creating a Java applet, which class must your applet extend?

A) java.awt.Frame
B) java.lang.Object
C) javax.swing.JApplet
D) java.applet.Applet

M1-Q27

Q2. What is the purpose of the codebase attribute in an <applet> tag?

A) Specifies Java source file name
B) Mandatory attribute for compiled code
C) Alternate text if browser can't run applet
D) Specifies base URL/directory of applet's class file

M2-Q14

Q3. Which HTML tag is used to embed a Java applet in a web page?

A) <java>
B) <script>
C) <embed>

D) <applet>

M2-Q24

Q4. Key difference between AWT component and Swing component?

A) AWT lightweight, Swing heavyweight
B) Swing component names often prefixed with 'J'
C) AWT supports more components
D) Swing platform-dependent; AWT not

M2-Q30

Q5. Why does an applet not have a main() method?

A) Applets outdated
B) main() replaced by paint()
C) Executed by browser/appletviewer controlling lifecycle via init(), start()
D) main() declared in Applet superclass and inherited

M3-Q18

Q6. How is a Java Applet program typically executed?

A) From command line using 'java'
B) Embedded in HTML, viewed via browser or appletviewer
C) As background OS service
D) Converted to native executable

M3-Q25

Q7. Correct way to pass 'username'='guest' from HTML to applet?

A) <applet code='...' username='guest'>
B) <applet code='...'><input name='username' ...></applet>
C) <applet code='...'><param name='username' value='guest'></applet>
D) Applets cannot receive parameters

M4-Q18

Q8. How must a Java Applet be executed after compilation?

A) Run java command on .class from console
B) Embed .class in HTML and load in browser or applet viewer
C) Create executable using 'appletc' compiler
D) Place .class in system directory

M4-Q30

Q9. How can you pass a color name from HTML to applet?

A) Command line arguments when launching applet viewer
B) Applet reads from config file automatically
C) Using <param> tag within <applet> tag in HTML
D) Must be hardcoded in Java source

M5-Q18

Q10. Which two packages are essential for basic Java applet programming?

A) java.lang and java.io
B) java.applet and java.awt
C) javax.swing and java.net
D) java.util and java.sql

M5-Q29

Q11. What is the typical behavior of a modern browser encountering an <applet> tag?

A) Downloads plugin automatically

B) Shows error and stops rendering
C) Ignores the applet tag entirely
D) Runs in secure sandbox mode

M5-Q30

Q12. If Point is declared as public static nested class inside Circle, how to create instance from another class?

A) new Point();
B) new Circle.Point();
C) Circle c = new Circle(); new c.Point();
D) Cannot be instantiated outside Circle

■ Answer Key — Applet Programming (AWT & Swing)

Q1 (M1-Q19): D	Q2 (M1-Q27): D	Q3 (M2-Q14): D
Q4 (M2-Q24): B	Q5 (M2-Q30): C	Q6 (M3-Q18): B
Q7 (M3-Q25): C	Q8 (M4-Q18): B	Q9 (M4-Q30): C
Q10 (M5-Q18): B	Q11 (M5-Q29): C	Q12 (M5-Q30): B

Topic 16: Output Tracing & Operator Questions

M1-Q4

Q1. Analyze the Java code with try-finally. What is the output? [InsideTry / InsideTryFinally / CaughtFinally / Compilation Error]

A) InsideTry
B) InsideTryFinally
C) CaughtFinally
D) Compilation Error

M1-Q5

Q2. Output of the Java program involving arithmetic?

A) 17
B) 15
C) 14
D) 12

M1-Q8

Q3. Output considering octal literals and char-to-int promotion?

A) 88
B) 810
C) 18
D) 64

M2-Q4

Q4. int i=10; int n=i++%5; int m=++i%5; Output of i+' '+n+' '+m?

A) 12 0 2
B) 11 0 1
C) 12 1 2
D) 11 1 1

M2-Q7

Q5. char[] letters={'n','p','t','e','l'}; print("" + letters[1] + letters[3]). Output?

A) 212
B) pe
C) Error
D) pt

M3-Q5

Q6. int x=5; y=10; z=--y+x++; print(x+y+z). Output?

A) 28
B) 29
C) 30
D) 31

M3-Q9

Q7. Integer i1=127, i2=127; print(i1==i2); Integer i3=128, i4=128; print(i3==i4). Output?

A) truetrue
B) falsefalse
C) truefalse
D) falsetrue

M4-Q4

Q8. Analyze the code snippet with post/pre-increment. Output?

A) 12
B) 10
C) 11
D) Compilation Error

M4-Q8

Q9. Output of the Java program with char/int arithmetic?

A) A82
B) 752
C) 732
D) Compilation Error

M5-Q7

Q10. Output of post/pre increment code snippet?

A) 5, 5
B) 6, 5
C) 5, 6
D) 6, 6

M5-Q9

Q11. Output of the following Java program?

A) true
B) false
C) Compilation error
D) ClassCastException

■ Answer Key — Output Tracing & Operator Questions

Q1 (M1-Q4): C	Q2 (M1-Q5): A	Q3 (M1-Q8): D
Q4 (M2-Q4): A	Q5 (M2-Q7): B	Q6 (M3-Q5): B
Q7 (M3-Q9): C	Q8 (M4-Q4): B	Q9 (M4-Q8): C
Q10 (M5-Q7): B	Q11 (M5-Q9): B	